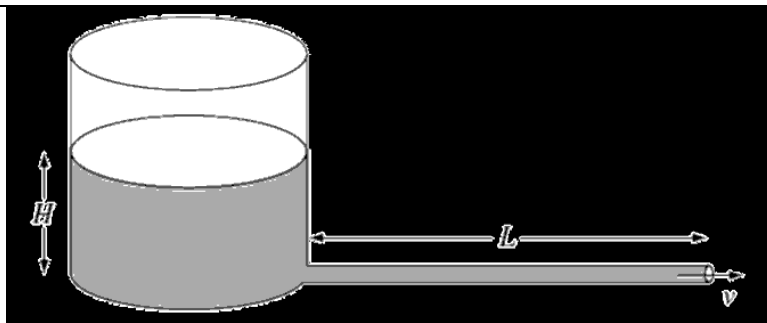


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Semester Examination – Summer 2022****Course: B. Tech.****Branch : Mechanical Engg.****Semester : IV****Subject Code & Name: BTMEC404 – Numerical Methods in Mechanical Engineering****Max Marks: 60****Date:****Duration: 3.00 Hrs.**

	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Explain the terms with the help of suitable examples: 1. True error 2. Approximate error 3. Truncation error 4. Round-off error	CO-1	06
B)	Write short note on 'Total numerical error'	CO-1	06
C)	Explain the terms: 1. Blunders 2. Formulation errors 3. Data uncertainty	CO-1	06
Q.2	Solve Any Two of the following.		
A)	Using bisection method find the roots of the equation $f(x) = x^{10} - 1$ Calculate the roots till the approximate error falls below 5%. Note that the root lies between 0 to 1.3.	CO-2	06
B)	Using Newton-Raphson method, find the root of the equation using initial guess as 0. $f(x) = e^{-x} - x$ Carry out minimum four iterations.	CO-2	06
C)	The velocity of water v (m/s), during the discharge from a cylindrical tank through a long pipe can be calculated as $v = \sqrt{2gH} \tanh\left(\frac{\sqrt{2gH}}{2L} t\right)$	CO-4	06



If $g = 9.81 \text{ m/s}^2$, H = initial head (m), L = Length of pipe (m), and t = elapsed time (s). Determine the head needed to achieve $v=5 \text{ m/s}$ in 2.5 sec for a 4 m long pipe. Use bisection method with initial guesses as 0 m and 2 m. carry out minimum three iterations.

Q. 3	Solve Any Two of the following.																
A)	Solve the following system of equations using Cramer’s rule. $2x_1 + x_2 + x_3 = 9$ $3x_1 + x_2 = 10$ $2x_2 + 5x_3 = 9$	CO-2	06														
B)	Solve the following system of equations using Gauss elimination method. $10x_1 + 2x_2 - x_3 = 27$ $-3x_1 - 6x_2 + 2x_3 = -61.5$ $x_1 + x_2 + 5x_3 = -21.5$	CO-2	06														
C)	Solve the following system of equations using Gauss elimination method with partial pivoting. $8x_1 + 2x_2 - 2x_3 = -2$ $10x_1 + 2x_2 + 4x_3 = 4$ $12x_1 + 2x_2 + 2x_3 = 6$	CO-2	06														
Q.4	Solve Any Two of the following.																
A)	Evaluate the following integral using Simpson’s 1/3 rd Rule. $\int_0^{\pi/2} (6 + 3 \cos x)dx$	CO-2	06														
B)	Evaluate the integral of the following tabulated data using trapezoidal rule. <table border="1"><tr><td>x</td><td>0</td><td>0.1</td><td>0.2</td><td>0.3</td><td>0.4</td><td>0.5</td></tr><tr><td>F(x)</td><td>1</td><td>8</td><td>4</td><td>3.5</td><td>5</td><td>1</td></tr></table> Also comment on how the accuracy of trapezoidal rule be increased?	x	0	0.1	0.2	0.3	0.4	0.5	F(x)	1	8	4	3.5	5	1	CO-2	06
x	0	0.1	0.2	0.3	0.4	0.5											
F(x)	1	8	4	3.5	5	1											
C)	Write algorithm for evaluating an integration using Trapezoidal rule.	CO-4	06														
Q. 5	Solve Any Two of the following.																
A)	Fit a straight line with zero intercept to the following data using least-square regression method.	CO-2	06														

	<table><tr><td>X</td><td>2</td><td>4</td><td>6</td><td>7</td><td>10</td><td>11</td><td>14</td><td>17</td><td>20</td></tr><tr><td>Y</td><td>1</td><td>2</td><td>5</td><td>2</td><td>8</td><td>7</td><td>6</td><td>9</td><td>12</td></tr></table>	X	2	4	6	7	10	11	14	17	20	Y	1	2	5	2	8	7	6	9	12		
X	2	4	6	7	10	11	14	17	20														
Y	1	2	5	2	8	7	6	9	12														
B)	Use Newton’s interpolating polynomial to determine y at x =8 to the best of accuracy using the following data. <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>5.5</td><td>11</td><td>13</td><td>16</td><td>18</td></tr><tr><td>Y</td><td>0.5</td><td>3.134</td><td>5.3</td><td>9.9</td><td>10.2</td><td>9.35</td><td>7.2</td><td>6.2</td></tr></table>	X	0	1	2	5.5	11	13	16	18	Y	0.5	3.134	5.3	9.9	10.2	9.35	7.2	6.2	CO-2	06		
X	0	1	2	5.5	11	13	16	18															
Y	0.5	3.134	5.3	9.9	10.2	9.35	7.2	6.2															
C)	Solve the following from $t = 1.5$ to $t = 2.5$ suing fourth-order Runge-Kutta method. $\frac{dy}{dx} = \frac{-2y}{1 + t}$ If $y(0) = 2$	CO-2	06																				
	*** End ***																						